

- Functional Requirements Document (FRD) for Data Integration and Aggregation Application**
- Introduction:**

The purpose of this FRD is to outline the functional requirements for the data integration and aggregation application. The application is designed to ingest customer and order data from CSV files, perform data cleansing, create summary tables, and implement Slowly Changing Dimension (SCD) type 2 logic to track changes in customer data.

- Functional Requirements:**

1. **Data Ingestion**

- The system shall read CSV data from the specified volumes (`/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``).
- The system shall load the ingested data into Delta tables (``customer`` and ``order``).

2. **Data Schema**

- The system shall create the ``customer`` table with the schema: ``CustId`, `Name`, `EmailId`, `Region``.
- The system shall create the ``order`` table with the schema: ``OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId``.

3. **Data Cleansing**

- The system shall remove "Null" or null records from both ``customer`` and ``order`` tables.
- The system shall remove duplicate records from both ``customer`` and ``order`` tables.

4. **Order Summary Table Creation**

- The system shall create the ``ordersummary`` table if it does not exist in the ``gen_ai_poc_databrickscoe`` catalog and ``sdlc_wizard`` schema.
- The schema for the ``ordersummary`` table shall be: ``CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `StartDate`, `EndDate`, `IsActive``.

5. **Data Integration and SCD Type 2 Implementation**

- The system shall join ``customer`` and ``order`` data using the ``CustId`` field.
- The system shall load the joined data into the ``ordersummary`` table.
- The system shall implement SCD type 2 logic to update the ``ordersummary`` table whenever there is a change in the ``customer`` table.
- Old records shall be made inactive, and new records shall be made active.
- ``StartDate`` and ``EndDate`` shall be updated accordingly.

6. **Customer Aggregate Spend Table Creation**

- The system shall create the `customeraggregatespend` table if it does not exist in the `gen\_ai\_poc\_databrickscoe` catalog and `sdic\_wizard` schema.
- The schema for the `customeraggregatespend` table shall be: `Name`, `TotalAmount`, `Date`.

7. **Data Aggregation**

- The system shall aggregate the `TotalAmount` column from the `ordersummary` table and group by `Name` and `Date` columns.
- The system shall load the aggregated data into the `customeraggregatespend` table.

• Data Flow Diagram

• System Components:

1. Data Ingestion Module
2. Data Cleansing Module
3. Order Summary Table Creation Module
4. SCD Type 2 Implementation Module
5. Customer Aggregate Spend Table Creation Module
6. Data Aggregation Module

• Interface Requirements:

1. The system shall read CSV data from the specified volumes.
2. The system shall load data into Delta tables.
3. The system shall create the required tables (`ordersummary` and `customeraggregatespend`) if they do not exist.

• Performance Requirements:

1. The system shall ensure data consistency and integrity.
2. The system shall be able to handle large volumes of data.
3. The system shall be scalable and flexible to accommodate changing business needs.

By following this FRD, the development team should be able to design and implement a data integration and aggregation process that meets the business requirements and is scalable, flexible, and maintainable.